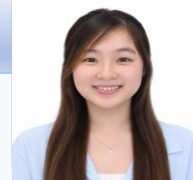


HIDDEN IN SILICONE: DIAGNOSTIC CHALLENGES OF A BORDERLINE PHYLLODES OBSCURED BY FREE SILICONE BREAST INJECTION

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BACKGROUND & AIM

- Phyllodes tumours are rare fibroepithelial neoplasms accounting for less than 1% of all breast tumours (1). Aetiology is not well elucidated, and only risk factor is Li-Fraumeni's Syndrome (3).
- Imaging features overlap with fibroadenomas and some breast cancers
- We report a case demonstrating the diagnostic challenges of heavily infarcted phyllodes tumour in the setting of extensive silicone-related changes and superimposed breast infection.

CASE PRESENTATION

- 53-year-old Filipino woman, with previous bilateral free silicone breast injections.
- P/w bilateral palpable firm breast masses and pain
- Underwent breast imaging evaluation

IMAGING

BILATERAL MAMMOGRAM - PERFORMED 3 YEARS PRIOR

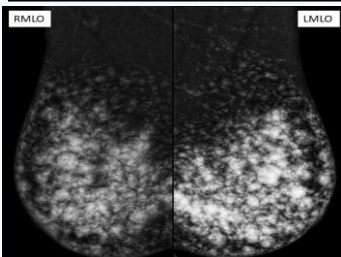


Fig 1: MG confirms diffuse hyperdense opacities from free silicone injection augmentation throughout both breasts, severely limiting MG assessment.

Only MRI imaging was feasible (below).

MRI BREASTS:

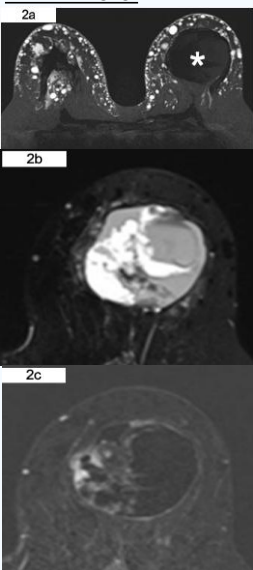
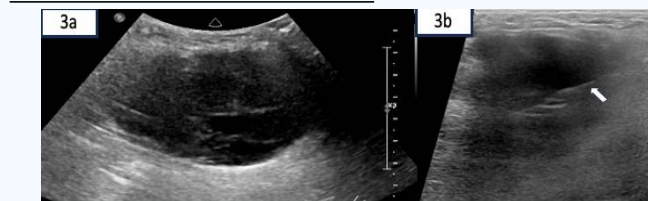


Fig 2a: T2w TIRM water saturation, silicone-select sequence shows multiple silicone granulomas and free silicone globules in both breasts. A large 8.5 cm circumscribed mass is seen in the left upper breast (*), which does not contain silicone signal and is therefore not a silicone granuloma.

Fig 2b : TIRM silicone saturated, water select sequence shows loculated components containing hyperintense water signal within the mass. Other loculated areas of isointense signal are hyperintense on T1w FSE sequence (not shown) - probably related to blood products .

Fig 2c: T1w FL3D SPAIR post-contrast subtracted image shows a well-defined, thin-walled, mass with heterogeneous enhancement of the solid component with restricted diffusion on DWI (not shown). The mass was deemed suspicious.

ULTRASOUND & CORE NEEDLE BIOPSY



US breast was subsequently performed to assess sonographic visibility of the mass for ultrasound-guided core biopsy.

Fig 3a: Classical "snowstorm" appearance in the breast parenchyma due to diffuse silicone granulomas with posterior acoustic shadowing. Large complex solid-cystic mass within the central left breast was partially visualised

Fig 3b: US-guided core needle biopsy was performed (Arrow). Thick brown fluid was also aspirated from the biopsy tract.

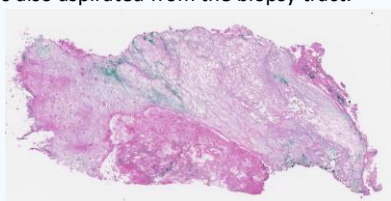
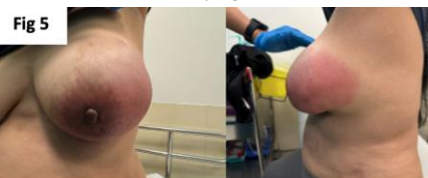


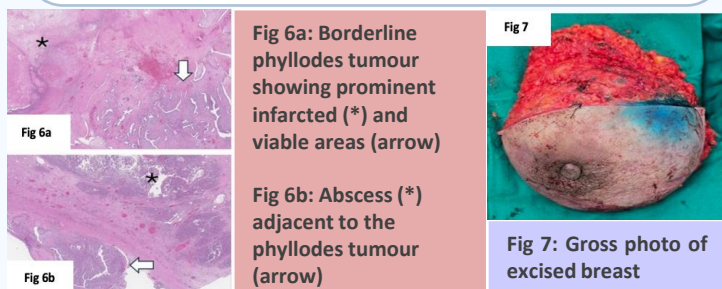
Fig 4: Core biopsy showed areas of infarcted breast tissue with no malignancy seen in the viable tissue

SUBSEQUENT MANAGEMENT

- Given the non-diagnostic nature of the core biopsy, patient was offered surgical biopsy to establish definitive diagnosis
- Patient was also keen for breast surgery due to persistent pain
- Counselling for Nipple Sparing Mastectomy with delayed reconstruction in a few months' time.
- However, the patient re-presented 3 months later with progressively worsening breast pain, diffuse erythema involving the lateral breast & nipple discharge. Clinical diagnosis was infected siliconomas or underlying abscess formation (Fig 5).



- Simple mastectomy performed immediately due to ongoing infection
- Histology: **borderline phyllodes tumours + silicone mastitis**
- Uneventful recovery; advised future MRI screening



DISCUSSION

Impact of Silicone on Diagnostic Accuracy

- Free silicone injection augmentation is common in Asia
- The free silicone deposits a/w granulomatous reaction cause significant masking and sonographic impairment on MG and US
- MRI is required for breast assessment in such cases

Differential Diagnosis of Solid- Cystic Masses

- All solid-cystic masses need to be evaluated to exclude malignancy
- Important differentials include papillary cancers, high grade carcinomas with necrosis, and phyllodes tumours (occasionally)
- Tissue diagnosis is therefore recommended
- Other differentials include galactoceles (in lactating breasts) and abscess (clinical and radiological signs of inflammation)

The "Infarction Trap" in Core Needle Biopsies

- Rapidly growing tumours such as phyllodes tumour, often outstrip their vascular supply, leading to central necrosis
- Core needle biopsy may only sample the necrotic zones
- Presence of infarcted tissue without any viable cells on histology should undergo excision biopsy for definitive histological diagnosis and to exclude malignancy
- If obscured on MMG / US, MRI-guided localisation and excision biopsy may be necessary
- Emerging evidence of contrast-enhanced mammography-guided vacuum-assisted biopsy in silicone-injected breasts, as an alternative to MRI-guided biopsies (5)

CONCLUSION & LEARNING POINTS

- Free silicone breast injections pose significant challenges in the detection and diagnosis of breast malignancies.
- MRI plays a critical role in lesion detection and diagnostic work-up in such cases
- Be aware of the "infarction trap" of core biopsies, especially in solid-cystic masses. In cases of radiological and histological discordance or non-diagnostic needle biopsy, excision biopsy is recommended

REFERENCES

- (1) Zhou ZR et al. Phyllodes tumors of the breast: diagnosis, treatment and prognostic factors related to recurrence. J Thorac Dis. 2016 Nov;8(11):3361-8.
- (2) Salvadori B et al. Surgical treatment of phyllodes tumors of the breast. Cancer. 1989 Jun 15;63(12):2532-6.
- (3) Birch JM et al. Relative frequency and morphology of cancers in carriers of germline TP53 mutations. Oncogene. 2001;20(34):4621-8. doi:10.1038/sj.onc.1204621
- (4) Bhardwaj P et al. Free Silicone Injections to the Breast: Delayed Complications and Surgical Management of Sequelae. Plast Reconstr Surg Glob Open [Internet]. 2020;8(11).
- (5) Cheung YC et al. A case report of breast cancer in silicone-injected breasts diagnosed by an emerging technique of contrast-enhanced mammography-guided biopsy. Front Oncol. 2022 Jul 22;12.